

Fluidic-Embedded TIM for SiC Power Electronics

Executive Project Outline — Team Roles, Tools & 10-Month Timeline
(Phase One: Simulation)

August 2026

1. Project Scope

Design-space simulation of a fluidic-embedded thermal interface material (TIM) for SiC power modules, combining **MLIP-based molecular dynamics (MD)** and **conjugate heat transfer CFD**. Ten-month timeline, four-person team, results to be handed down to juniors as the team progresses.

Neither MD nor CFD are tools the team has deep prior experience with, so the timeline is deliberately structured around a **learning curve**: get a basic, working model in each domain first, validate it against known/textbook results, and only then progressively add the complexity that makes the model project-specific (multi-material interfaces, full geometry, coupled MD-derived boundary conditions). Complexity is layered in, not attempted all at once.

Fabrication (Phase Two) is explicitly out of scope for this phase.

Two components feed the same design map:

- **MD** — derives material inputs (bulk thermal conductivity of the TIM matrix, interfacial/Kapitza thermal resistance at the TIM-metal boundary) that would otherwise be guessed from datasheet ranges.
 - **CFD** — conjugate heat transfer simulation of the full stack (die → DBC → baseplate → fluidic TIM → heat sink), parametrically swept across channel geometry and bond-line thickness, using the MD-derived properties as inputs.
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2. Team Roles

Person A — MD: Bulk Conductivity

Sets up pretrained MLIP calculators for the candidate TIM matrix materials and runs simulations to extract bulk thermal conductivity as a function of filler loading and filler geometry.

- **Matrix candidates**: filled silicone elastomer, metal-filled polymer composite (per the material shortlist in Section 7 of the project report)
- **Model**: pretrained foundation MLIP suited to organic/molecular systems (e.g. MACE-OFF) for the polymer matrix; a pretrained foundation MLIP suited to inorganic/crystalline systems (e.g. MACE-MP-0, CHGNet) for the metal filler phase
- **Method**: Green-Kubo (equilibrium MD) or HNEMD, run via ASE or GPUMD
- **Validation step**: benchmark predicted k against known

experimental PDMS/silicone conductivity ($\sim 0.15\text{--}0.27\text{ W/m}\cdot\text{K}$)
before trusting composite-loading results

Person B — MD: Interfacial Resistance

Builds the TIM-metal interface cell (matrix against Cu baseplate / Al or Cu heat sink) and runs NEMD to extract interfacial thermal conductance (Kapitza resistance).

- **Model:** same pretrained foundation MLIPs as Person A, applied across the heterostructure interface
- **Method:** NEMD with local thermostat regions (temperature-profile method), via ASE or GPUMD
- **Known limitation:** pretrained foundation models are not trained specifically on matrix-filler or matrix-metal interface chemistry — cross-validate against literature Kapitza-resistance ranges for comparable polymer-metal pairs and treat results as a bounding estimate rather than a precise value. Document this explicitly as a risk item.

Person C — CFD: Geometry, Meshing & Parametric Sweep

Builds the parametric stack geometry and runs the core conjugate heat transfer sweep.

- **Geometry:** SiC die, die-attach, DBC substrate, Cu baseplate, fluidic-embedded TIM (channel width/pitch as design variables), heat sink
- **Mesh:** independence study before any sweep results are trusted
- **Physics:** conjugate heat transfer — solid conduction + laminar microchannel flow, steady-state and transient (switching-frequency thermal transients)
- **Sweep parameters:** channel width (10–200 μm), pitch (50–500 μm), bond-line thickness (50–500 μm), coolant flow rate (10–500 mL/min)
- **Inputs from MD:** matrix thermal conductivity (Person A) and interfacial conductance boundary condition (Person B) replace literature-range placeholders
- **Tools:** COMSOL Multiphysics (Heat Transfer + CFD modules) or OpenFOAM (chtMultiRegionFoam) as an open-source fallback

Person D — CFD: Benchmarking, Integration & Documentation

Runs the baseline comparison cases and owns turning the team's combined output into the final design map and report.

- **Baselines:** thermal grease, phase-change TIM, and indium foil TIM, each paired with a separate solid microchannel coldplate, using identical total thermal-path length and coolant conditions
- **Compliance calc:** analytical CTE-mismatch shear stress estimate across the same geometry sweep
- **MD → CFD handoff:** owns the data contract — a simple lookup table or fitted curve (k vs. filler %, G vs. interface pair), not raw trajectory data
- **Deliverables:** bond-line-thickness-vs-channel-geometry design map, R_{th} and pressure-drop plots, compliance-vs-rigidity overlay, final write-up
- **Tools:** Python (pandas/matplotlib) or MATLAB for post-processing, LaTeX/Overleaf for the report

3. Ten-Month Timeline

The timeline is structured in four stages: **Learn** → **Basic Model** → **Add Complexity** → **Integrate & Finalize**. Each stage ends with a checkpoint the whole team reviews together before moving on, so juniors coming in later can see exactly what “done” looked like at each level of complexity.

Stage 1 — Learning Curve (Months 1-2)

Goal: everyone can run a minimal, textbook-style version of their tool end-to-end before touching the real project geometry or chemistry.

Month	MD Track (A, B)	CFD Track (C, D)
1	Learn ASE/GPUMD basics; run a pretrained foundation MLIP (e.g. MACE-MP-0) on a simple known system (bulk Cu or bulk Si) and reproduce a textbook property (lattice constant, known k)	Learn COMSOL/OpenFOAM basics on a simple single-block conduction problem; reproduce a hand-calculable steady-state result
2	Run a first Green-Kubo/HNEMD calculation on a <i>simple</i> single-component matrix (e.g. bulk PDMS, no filler) and sanity-check against literature k	Build a simplified 2D single-channel version of the TIM geometry; get a first (unvalidated) conjugate heat transfer solution running

Checkpoint: everyone has one working, sanity-checked “toy” result in their domain.

Stage 2 — Basic Project Model (Months 3-5)

Goal: a first pass at the *actual* project system, deliberately simplified — single matrix material, single filler loading, 2D or coarse 3D geometry, no interfacial fine detail yet.

Month	MD Track (A, B)	CFD Track (C, D)
3	Select final pretrained MLIPs per domain (organic for matrix, inorganic for filler); benchmark both against known reference values	Finalize full 3D parametric geometry (die → DBC → baseplate → TIM → heat sink); mesh independence study
4	Run bulk- k for one matrix + one filler loading (A); build first simplified	Finalize boundary conditions from published SiC module data; run first full-stack case at

	matrix-metal interface cell (B)	one fixed geometry, literature-range material values as placeholder
5	Run first NEMD pass on the simplified interface (B); cross- check bulk k against Stage 1 toy result (A)	Run baseline cases (grease / PCM / indium + separate coldplate) at the same fixed geometry

Checkpoint: one complete, if simplified, thermal resistance number for the fluidic TIM concept vs. one baseline — the “does this basically work” result.

Stage 3 — Adding Complexity (Months 6-8)

Goal: layer in what Stage 2 deliberately left out — the parameter sweep, multiple matrix/filler combinations, and the harder interface chemistry.

Month	MD Track (A, B)	CFD Track (C, D)
6	Extend bulk-k runs across the full filler- loading range (A); refine interfacial NEMD, cross- validate against literature Kapitza- resistance ranges for comparable polymer- metal pairs (B)	Begin full parametric sweep: channel width (10–200 μm), pitch (50–500 μm), bond-line thickness (50–500 μm)
7	Finalize k(filler) curves and interfacial G values across all candidate matrices (A, B)	Continue sweep; feed in first round of real MD-derived properties in place of placeholders
8	Support/re-run edge cases flagged by the CFD sweep; document the interface-accuracy limitation explicitly	Re-run full sweep using finalized MD- derived k and interfacial boundary conditions; analytical CTE-mismatch compliance calc across the same grid

Checkpoint: full design map inputs exist for every geometry/material combination in scope.

Stage 4 — Integration & Finalization (Months 9-10)

Month	MD Track (A, B)	CFD Track (C, D)
9	Contribute MD methods section; support any final re- runs	Assemble bond-line- thickness-vs-channel- geometry design map; overlay compliance-vs- rigidity map

10	Final review pass on all MD data and limitations	Final write-up, cross-check plots against every sub-team's raw numbers, prepare handoff materials for juniors
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4. Combining Results at the End

All sweep runs must output to a **common schema** from Stage 1 — channel width, pitch, bond-line thickness, matrix identity, R_{th} , pressure drop, shear stress — into one shared spreadsheet or CSV, not four separate notebooks reconciled in month ten. Because the project is deliberately built up in stages of increasing complexity, this schema needs to stay stable *across stages* too: the Stage 2 “basic model” row format should be a subset of the Stage 3 “full sweep” row format, not a different structure that needs re-work later. The thermal design map and the compliance map are the same parameter grid viewed two ways, so the geometry and materials sweeps need to share identical parameter values to overlay cleanly.

Practically:

- Shared Git repo or drive folder with a locked column/naming convention decided at the start of Stage 1, before the first toy result is even logged
 - Doc lead (Person D) does a sanity merge at every stage checkpoint (end of months 2, 5, 8, 10), not just at the very end
 - MD → CFD handoff format agreed at the start of Stage 2: a simple fitted function or lookup table, never raw trajectory data
 - Each stage checkpoint is a good moment to bring juniors up to speed, since the model at that point is simpler and easier to explain than the final version
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5. Open Items / Risks to Track

- **Pretrained MLIP accuracy at the matrix-filler/matrix-metal interface is unvalidated** — cross-check against literature Kapitza-resistance ranges; report as a bounding estimate, not a precise result
- COMSOL license seat availability — check institute access before Stage 1 begins
- GPU/cluster access for MD runs — check alongside the COMSOL check
- Whether to run thermal-conductivity MD via ASE (flexible, Python-native) or GPUMD (faster, purpose-built for HNEMD) — decision deferred, revisit before Stage 2 begins
- Learning-curve risk: if Stage 1 toy results aren't converging by end of month 2, build in a buffer before committing to Stage 2's basic model rather than pushing forward on an unvalidated foundation